SMART INDIA HACKATHON 2025

TITLE PAGE

- **Problem Statement ID - SIH25024**
- **Problem Statement Title- Comprehensive Cloud-Based Practice Management & Nutrient Analysis Software for Ayurvedic Dietitians, Tailored for Ayurveda-Focused Diet Plans**
- **Theme- MedTech / Bio Tech / HealthTech**
- **PS Category- Software**
- **Team ID- 106873**
- **Team Name – Techh Titann**

TECHH TITANN

IDEA TITLE

SMART INDIA HACKATHON 2025

Proposed Solution (Described Idea/Solution/Prototype)

The **Ayurvedic Diet Management Platform** is an **AI-enabled solution** that helps practitioners create and manage **personalized diet charts** based on **Ayurvedic principles**. It includes a **dynamic database** of **8,000+ food items** with **scientifically calculated nutrients** tailored by **climate** and **region**.

The system enables **automated diet plan generation** aligned with **rasa, virya, and dosha balance**, and provides **dashboards** for **patient demographics**, **health parameters**, and **recipe-based** nutrient analysis. With **robust data security**, **HIS/mobile integration**, and **customizable workflows**, the platform ensures **efficient** and **holistic dietary care delivery**.

Unique Value Propositions (UVP)

CHALLENGES ADDRESSED

1. Validating 8000+ Food Items
2. Lack of Personalized Diet Plans
3. Supporting Regional Recipes & Seasonal variation
4. Balancing Dosha Needs & Modern Nutrition
5. Ensuring Patient Data Privacy
6. Practitioners lack real-time multi-patient monitoring
7. Convincing Traditional Healers

SOLUTIONS

1. Expert-Vetted Database: Database vetted by senior BAMS/MD experts with automated checks that flag inconsistent ingredient classifications
2. Personalized Plans: Layer a rule-based dosha engine on patient parameters (age, habits) with one-tap clinician overrides and reusable templates

5. Security by Design (SBD): Top-grade encryption, role-based access controls, and early legal compliance checks for regulations.
6. Personalized Patient Dashboard: Holistic patient view for progress, and feedback so practitioners spot trends and adjust plans quickly.

3. Regional Food Intelligence: Integrates seasonal availability, most potent foods aligned with Ayurvedic seasonal eating principles
4. Dosha-aware Nutrient Optimization: Using a constrained optimizer to enforce dosha rules and meet nutrient targets with dosha-appropriate swaps.

7. Localized low-friction clinic workflow: Provide regional recipes and seasonal swaps with a multilingual UI, and enable staff-assisted data entry, paper-to-digital scanning, and one-tap clinician review for fast, culturally relevant adoption.

TECHH TITANN

TECHNICAL APPROACH

SMART INDIA HACKATHON 2025

Tech stack and methodology

1. Build **responsive Frontend** with **React, Tailwind CSS, Material Ui & Flutter** for mobile-first design.
2. Secure **Node.js** and **Express.js REST API backend** with versioned endpoints.
3. Use **MySQL (RDS)** for relational data and **MongoDB(Atlas)** for semi structured metadata.
4. **JWT** sessions, **OAuth2.0 SSO**, and **RBAC** for authentication and authorization.
5. **Python ML service** with **scikit-learn, TensorFlow**, or **PyTorch** for **nutrient calculations**.
6. **MLOps pipeline** using **SageMaker** or **MLflow integrated** with a **feature store**.
7. **Knowledge graph database** for comprehensive, detailed **Ayurvedic food attribute representation.**
8. **SQL** full-text search combined with **AI** recommendations via a **PyTorch NLP pipeline.**
9. **Blockchain integration** with **Hyperledger Fabric** or **Ethereum** for **reputation** and **contracts.**
10. **Multilingual support** using **Translation APIs** from **Google** or **Microsoft** for **global reach.**
11. **Dockerized AWS deployment** with **CI/CD pipelines** and end-to-end **testing.**

Hardware Specifications

- standard hardware of computer required.

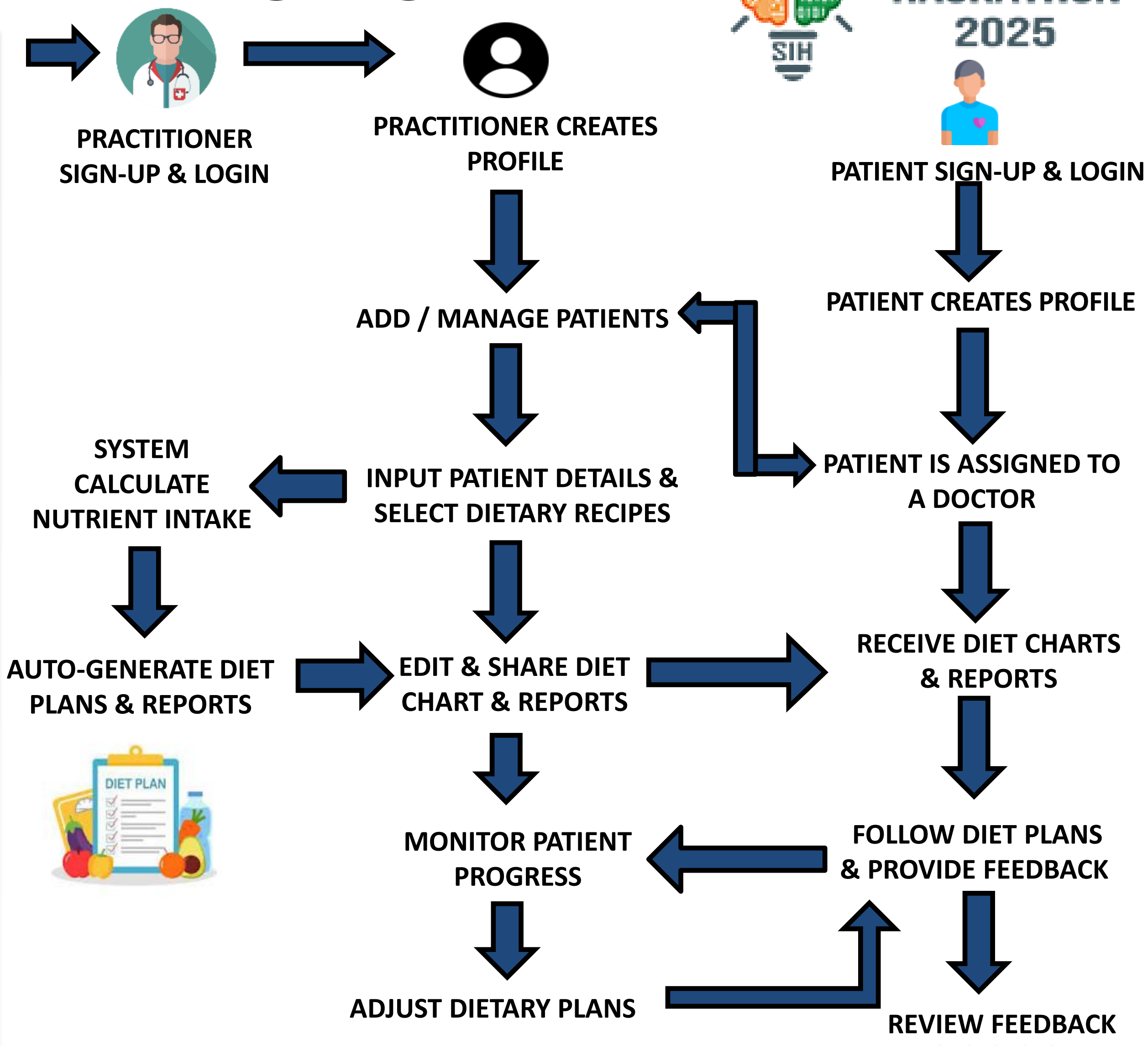

Fig.1.1 Work Flow Architecture of the Platform

TECHH TITANN

FEASIBILITY AND VIABILITY

SMART INDIA HACKATHON 2025

Technical Feasibility

- Development can be carried out using **standard hardware** and **open-source / free software frameworks.**
- Integration with **nutrient databases** and **Ayurvedic food categorization modules** can be achieved via APIs.

Schedule Feasibility

- **MVP** with **AI diet engine** and **patient module** deliverable in 9–10 months using agile sprints.
- **Continuous updates** aligned with evolving **Ayurvedic** and **nutrition research**.

Legal & Compliance Feasibility

- Compliant with **HIPAA, DISHA,** and Indian health data regulations to ensures **patient safety** and **confidentiality**.
- Aligns with **AYUSH guidelines** for Ayurveda practice.
- **Regular audits** and **certifications** will ensure **legal** and **ethical medical use**.

Market Feasibility

- Strong demand exists for **digitized Ayurvedic healthcare**, as manual prescriptions still dominate.
- Integrating **Indian food databases, Ayurvedic logic**, and **multilingual support** boosts adoption in urban and semi-urban clinics.
- Rising interest in **holistic care, personalized nutrition**, and **wellness apps** validates strong market potential.

PROBLEM 1

Currently, no government platform **integrates modern nutritional science with Ayurvedic principles**. Most private platforms follow **Western models** and **overlook Ayurvedic concepts** such as **rasa**, **virya**, and **dosha balance**.

SOLUTION 1

- Develop a government-led open digital platform that **merges modern nutrient science with Ayurveda, powered by AI.**
- Provide **APIs** for **developers**, **coaches**, and **healthcare apps**.
- Create a unified **food ontology combining Ayurvedic parameters with macronutrients** and **micronutrients**

PROBLEM 2

Ayurvedic practitioners still rely on **handwritten diet charts**, which are **time-consuming** and **inconsistent**.

SOLUTION 2

- **Digitize** handwritten diet **charts** using **OCR** and **AI automation**.
- Integrate with **EHR** and **cloud platforms** for seamless **data management**.
- Enable **easy export** and **sharing** of **diet charts** for practitioners and patients.

PROBLEM 3

Ayurvedic diet plans are **static** and **lack real-time health monitoring**, limiting personalization and **timely intervention**.

SOLUTION 3

- Integrate wearable **IoT devices** for **real-time monitoring**.
- Use **AI-powered** in-app **Nadi Pariksha** for **personalized insights**.
- Provide **real-time adaptive dashboards** and **alerts** for practitioners and patients.

TECHH TITANN

IMPACT AND BENEFITS

SMART INDIA HACKATHON 2025

IMPACT ON TARGET AUDIENCE

Nutritionist

- Automated generation of detailed, nutrient-analyzed diet charts.
- Blockchain & transparency :Nutritionists as well as patient data are secure and no third party involve.
- Enhances precision in patient-specific recommendations that enhance patient belief.
- Multilingual :Offer customer support in multiple languages. This can include live chat, email support, and comprehensive FAQs.
- Enhanced visibility : Ratings and reviews increase credibility and attract patients.
- Integrated progress tracking for outcome monitoring and easy to maintain patience health care.

Patient

- Delivers accessible, standardized Ayurvedic diet recommendations.
- Encourages data-ai-driven approaches in health research and patient care.
- Integrates ancient dietary knowledge with digital health resources and gives proper treatment.
- Expands reach of validated Ayurvedic guidance to non-specialist users.

BENEFIT OF SOLUTION

SOCIAL

- ❖ Digitization of patient records, diet charts, and medical parameters streamlines data management, minimizes human error, and ensures consistency in recommendations.
- ❖ effectively bridging the gap between traditional Ayurvedic knowledge and modern healthcare delivery systems.

ECONOMIC

- ❖ Automated patient management and electronic diet chart generation significantly reduce manual workload for Ayurveda practitioners.
- ❖ Health facilities—including clinics, wellness centers, and hospitals—can expand their service capacities, optimize workflow, and increase revenue streams.

CULTURAL

- ❖ Integrating ancient dietary frameworks with modern nutritional science increases both the relevance and accessibility of Ayurveda for contemporary populations.
- ❖ approach supports cultural heritage initiatives it promotes the use of seasonal, locally-sourced food, reinforcing alignment with region-specific dietary habits and sustainable resource utilization.

TECHH TITANN

RESEARCH AND REFERENCES

SMART INDIA HACKATHON 2025

1. **Annual Report 2024-25.** Ministry of AYUSH, Government of India, 2025.
   https://ayush.gov.in/resources/pdf/annualReport/Annual-Report24-25_English.pdf

2. **Ayurveda-Based Diet & Lifestyle Guidelines: Skin Diseases.** Central Council for Research in Ayurvedic Sciences, June 2024.
   https://ccras.nic.in/wp-content/uploads/2024/06/Ayurveda-Based-diet-lifeStyle-Guidelines-Skin-Diseases.pdf

3. ***Ayurveda Based Dietary and Lifestyle Advocacy for Mental Health & Psychiatric Disorders***. CCRAS, Ministry of AYUSH, Govt. of India, 2024.
   https://ccras.nic.in/wp-content/uploads/2024/07/Ayurveda-Based_Dietary_Guidelines_Mental_Disorders-1.pdf

4. **Department of Ayush.** *Traditional Food Recipes from AYUSH.* New Delhi: Directorate General of Health Services, Government of India, 2022.
   https://www.dghindia.gov.in/assets/downloads/635122af7ecbbTraditionalFoodRecipesFromAYUSH.pdf

5. ***Ayush Recommendations for the Public on Holistic Health & Well-Being: Preventive Measures and Care during COVID-19 & Long COVID-19.*** Ministry of Ayush, Government of India, 2022.
   https://ayush.gov.in/resources/pdf/health/ARPHHC19.pdf

6. ***Ayush Dietary Advisory for Kuposan Mukt Bharat.*** Ministry of AYUSH, Government of India, 2021.
   https://indianembassy-moscow.gov.in/pdf/jncc/Ayush-Dietary%20Advisory.pdf

7. ***Charaka Samhita***. [Book] .National Institute of Indian Medical Heritage, Government of India.

8. **Explanation Video for Presentation**
   https://youtu.be/5zy8_Oy84jQ?si=xvXGBmUszlidzLGz

9. **Prototype Demo Video**
   https://youtu.be/Xf5Z6COg8ws?si=QXqwkEeQAJttJwHt